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LANDMARK comparison of early outcomes of newer-generation Myval transcatheter heart valve series with contemporary valves (Sapien and Evolut) in real-world individuals with severe symptomatic native aortic stenosis: a randomised non-inferiority trial

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Summary

Background Transcatheter aortic valve implantation is an established, guideline-endorsed treatment for severe aortic stenosis. Precise sizing of the balloon-expandable Myval transcatheter heart valve (THV) series with the aortic annulus is facilitated by increasing its diameter in 1·5 mm increments, compared with the usual 3 mm increments in valve size. The LANDMARK trial aimed to show non-inferiority of the Myval THV series compared with the contemporary THVs Sapien Series (Edwards Lifesciences, Irvine, CA, USA) or Evolut Series (Medtronic, Minneapolis, MN, USA).

Methods In this prospective, multinational, randomised, open-label, non-inferiority trial across 31 hospitals in 16 countries (Germany, France, Sweden, the Netherlands, Italy, Spain, New Zealand, Portugal, Greece, Hungary, Poland, Slovakia, Slovenia, Croatia, Estonia, and Brazil), 768 participants with severe symptomatic native aortic stenosis were randomly assigned (1:1) to the Myval THV or a contemporary THV. Eligibility was primarily decided by the heart team in accordance with 2021 European Society of Cardiology guidelines. As per the criteria of the third Valve Academic Research Consortium, the primary endpoint at 30 days was a composite of all-cause mortality, all stroke, bleeding (types 3 and 4), acute kidney injury (stages 2–4), major vascular complications, moderate or severe prosthetic valve regurgitation, and conduction system disturbances resulting in a permanent pacemaker implantation. Non-inferiority of the study device was tested in the intention-to-treat population using a non-inferiority margin of 10·44% and assuming an event rate of 26·10%. This trial is registered with ClinicalTrials.gov, NCT04275726, and EudraCT, 2020-000137-40, and is closed to new participants.

Findings Between Jan 6, 2021, and Dec 5, 2023, 768 participants with severe symptomatic native aortic stenosis were randomly assigned, 384 to the Myval THV and 384 to a contemporary THV. 369 (48%) participants had their sex recorded as female, and 399 (52%) as male. The mean age of participants was 80·0 years (SD 5·7) for those treated with the Myval THV and 80·4 years (5·4) for those treated with a contemporary THV. Median Society of Thoracic Surgeons scores were the same in both groups (Myval 2·6% [IQR 1·7–4·0] *vs* contemporary 2·6% [1·7–4·0]). The primary endpoint showed non-inferiority of the Myval (25%) compared with contemporary THV (27%), with a risk difference of –2·3% (one-sided upper 95% CI 3·8, $p_{non\text{-}inferiority}$<0·0001). No significant difference was seen in individual components of the primary composite endpoint.

Interpretation In individuals with severe symptomatic native aortic stenosis, the Myval THV met its primary endpoint at 30 days.

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Introduction

Transcatheter aortic valve implantation (TAVI) is an established, minimally invasive procedure to treat severe aortic stenosis that has a Class I, Level A recommendation from the American College of Cardiology and American Heart Association and from the European Society of Cardiology and European Association for Cardio-Thoracic Surgery in older people

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Research in context

Evidence before this study

We searched PubMed from database inception to April 7, 2024, for earlier published evidence on Myval transcatheter heart valve (THV) series (Myval and Myval Octacor THV) for the treatment of native aortic stenosis, using the search term "Myval". There were 25 original research articles excluding animal studies, case reports, and review articles. Of these 25 articles, seven were comparative studies between Myval and other types of THV; four were retrospective studies without matching, and three were with propensity matching (one prospective and two retrospective). However, there were no randomised trials comparing Myval and other THVs.

Added value of this study

To our knowledge, the LANDMARK trial is the first randomised trial to compare Myval THV and contemporary THVs. The indication of transcatheter aortic valve implantation (TAVI) was judged by the local heart team and endpoints were adjudicated by an independent clinical event committee according to the third Valve Academic Research Consortium definition. Primary selection of participants by the heart team has expanded the indications of TAVI to those with complex anatomy, such as small aortic annuli and bicuspid aortic valves, as frequently encountered in clinical practice. The LANDMARK trial establishes the non-inferiority of Myval compared with contemporary THVs for early safety and effectiveness composite endpoints at 30 days for the treatment of individuals with symptomatic severe native aortic stenosis. Both groups had similar, statistically significant improvements in haemodynamic parameters, mean pressure gradients, and effective orifice areas between baseline and 30 days. In more than 48% of participants treated with Myval, the selection of intermediate sizes was based on the quantitative measurement of the aortic annulus area provided by preprocedural multislice CT.

Implications of all the available evidence

Since 2020, there have been three head-to-head comparisons of TAVI devices for aortic stenosis. In the SCOPE II trial, ACURATE neo failed to meet the prespecified criteria for non-inferiority to Evolut. In the PORTICO IDE trial, the Portico valve met non-inferiority to the contemporary valves (Sapien and Evolut series) in the primary endpoint, although Portico was associated with a higher 30-day safety endpoint. The SOLVE-TAVI randomised trial showed the equivalence of Evolut R and Sapien 3 in 447 individuals with severe aortic stenosis undergoing TAVI. In summary, Myval and Portico are non-inferior to two contemporary THVs for the treatment of severe aortic stenosis. The hypothetical benefit of the tailored sizing of Myval by availability of intermediate sizes needs to be investigated in future trials with a specific predefined endpoint or by superior clinical outcomes in the long term.

(aged ≥75 years), or in those who are considered at high risk (Society of Thoracic Surgeons [STS] Predicted Risk of Mortality or EuroSCORE II >8%) or unsuitable for surgery.[1–3] The indications of TAVI have expanded to select low-risk, and thereby relatively younger, individuals.[4–7]

Transcatheter heart valves (THVs) can be balloon-expandable valves (BEVs) or self-expanding valves (SEVs), with the latter offering the advantage of providing a larger effective orifice area (EOA) and lower gradients, particularly in valves with a supra-annular design.[8] However, compared with BEVs, these SEV bioprostheses have a higher risk of requiring permanent pacemaker implantation (PPI) and paravalvular leaks.[9] As the technology has evolved, several new-generation THVs have been designed to specifically incorporate novel features to mitigate procedural complications such as paravalvular leaks, vascular injuries, and conduction disturbances, thereby improving clinical outcomes.[10] The Myval THV is a novel approach with a BEV that has diversified the traditional sizing of existing BEVs and SEVs, which have increments of 3 mm in nominal valve size. The Myval THV enables precise and more appropriate matching of the device with the aortic annulus, with a 1·5 mm diameter increment between each nominal device size. The valve frame is manufactured from cobalt alloy metallic tube in 1·5 mm increments (eg, 20·0 mm, 21·5 mm, 23·0 mm) in order to cope with the wide range of aortic annulus dimensions (area from 270 mm^2 to 840 mm^2).[11] The safety and efficacy of the Myval THV in single-arm or propensity-matched studies have already been shown in high-risk, intermediate-risk, and low-risk individuals with severe symptomatic aortic stenosis and in bicuspid anatomy.[12–19]

Randomised controlled trials comparing TAVI devices[9,20–24] are currently limited to head-to-head comparisons of SEVs versus BEVs, or SEVs versus SEVs, except for the PORTICO IDE trial,[24] which compared the Portico SEV (Abbott, Green Oaks, IL, USA) with the Sapien BEV series (Edwards Lifesciences, Irvine, CA, USA) and the Evolut SEV series (Medtronic, Minneapolis, MN, USA).

The LANDMARK trial aimed to investigate non-inferiority of the Myval THV series compared with contemporary THVs (Sapien or Evolut series) in individuals with severe symptomatic aortic stenosis, with primary safety and effectiveness endpoints reported at 30 days according to the third Valve Academic Research Consortium (VARC-3), together with clinical and haemodynamic outcomes.[25,26]

Methods

Study design and participants

The LANDMARK trial is a prospective, randomised, multinational, multicentre, open-label, non-inferiority

trial conducted at 31 centres across Germany, France, Sweden, the Netherlands, Italy, Spain, New Zealand, Portugal, Greece, Hungary, Poland, Slovakia, Slovenia, Croatia, Estonia, and Brazil. The trial was designed to compare clinical outcomes of the Myval BEV series with contemporary THV systems in individuals with severe symptomatic native aortic stenosis with an indication for TAVI.

Full details of the trial design, as well as the amendment on eligibility that took place in October, 2021, are published elsewhere.[25,26] In brief, instead of the itemised and detailed eligibility criteria included in the initial protocol, the amended eligibility criteria made the local heart team responsible for assessing the indication for TAVI, thus enabling investigators to stay aligned with the latest guidelines for managing severe symptomatic aortic stenosis.[2] Adults (aged ≥18 years) with severe symptomatic native aortic stenosis who were deemed eligible by the local heart team to undergo TAVI with all three study devices were considered suitable for enrolment. Participants who were not willing to give informed consent were excluded. A pre-screening committee assessed the appropriateness of TAVI with the three different study valves, with the assessment of preprocedural multislice CT (MSCT) provided by the TAVI Core lab (India). Participants with an aortic annulus mean diameter exceeding 29 mm were excluded from randomisation but included in the XL Nested registry for extra-large diameter Myval (ie, 30·5 and 32 mm). The final decision for enrolment of the subject in the trial was taken by the site's heart team.

The study protocol (appendix 1), informed consent documents, and other relevant study documents were approved by the ethics committees of the respective study centres (appendix 2 pp 3–4). Written informed consent was obtained from all the study participants before sharing the screening documents to check their eligibility. The study was conducted in compliance with the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use standards for clinical research.

This trial is registered with ClinicalTrials.gov, NCT04275726, and EudraCT, 2020-000137-40.

Randomisation and masking

An interactive web response system was used to randomly assign eligible participants (1:1) to the Myval THV series (n=384) or contemporary THV series (n=384); minimisation and equal allocation were also performed for each valve in the contemporary THV group. A computer-generated randomisation code was maintained centrally, so no investigational site had any advance information about treatment allocation. The mechanism of contact was through a system-generated email for the particular site, and through this process any selection bias was avoided. A covariate-adaptive randomisation procedure was used to minimise imbalances of the covariate within each treatment group (STS Predicted Risk of Mortality version 2·9 [low risk <4%, intermediate risk 4–8%, and high risk >8%]), so asymptotically the power of testing the treatment effects would be greatest and the selection bias minimal. As this is an open-label trial, the treating physicians and participants were aware of the treatment allocation. The three types of THV series used in the study are described elsewhere.[4,25]

Procedures

Preprocedural assessment included physical examination, medical history, laboratory investigations, electrocardiogram (ECG), echocardiogram, and MSCT. After randomisation, the size of the device to be implanted was decided by the heart team based on the annular dimensions from the baseline MSCT. The transfemoral approach was mandated in the protocol. Device implantation was performed as per the instructions for use of the device. Despite cusp overlap being a traditional method for implantation, the technique for implantation was left to the investigator's discretion. Sedation or general anaesthesia was left to the discretion of the investigator. Predilatation and postdilatation were performed at the discretion of the operator to appropriately deploy the THV device. An aortogram was performed at the end of the procedure as per the standard protocol for offline evaluation of regurgitation fraction by videodensitometry.[27] Surgical or non-surgical closure of the femoral access was left to the discretion of the operators. As per the protocol, indications for postdilatation were not specified and it was left to the investigator's discretion and as per the routine practice at the centre. After TAVI, all participants were kept under observation to monitor clinical events according to local practice and had a postprocedural ECG, echocardiogram, and laboratory investigations. The antithrombotic strategy was recommended as per European Society of Cardiology and European Association for Cardio-Thoracic Surgery guidelines.[2] At 30-day follow-up, a clinical assessment, ECG, and echocardiogram were performed. The detailed study schedule and assessments are summarised elsewhere.[25]

Outcomes

The primary combined safety and effectiveness endpoint at 30 days was a composite of VARC-3 defined endpoints: all-cause mortality, all stroke, bleeding (VARC type 3 and 4), acute kidney injury (stages 2, 3, and 4), major vascular complications, moderate or severe prosthetic valve regurgitation, and conduction system disturbances resulting in a new PPI.[28,29] Secondary endpoints were the individual components of the primary endpoint, technical success, device success, and early safety endpoints at 30 days, as defined in VARC-3 and listed in the protocol. Functional improvement was assessed by change in New York Heart Association functional class

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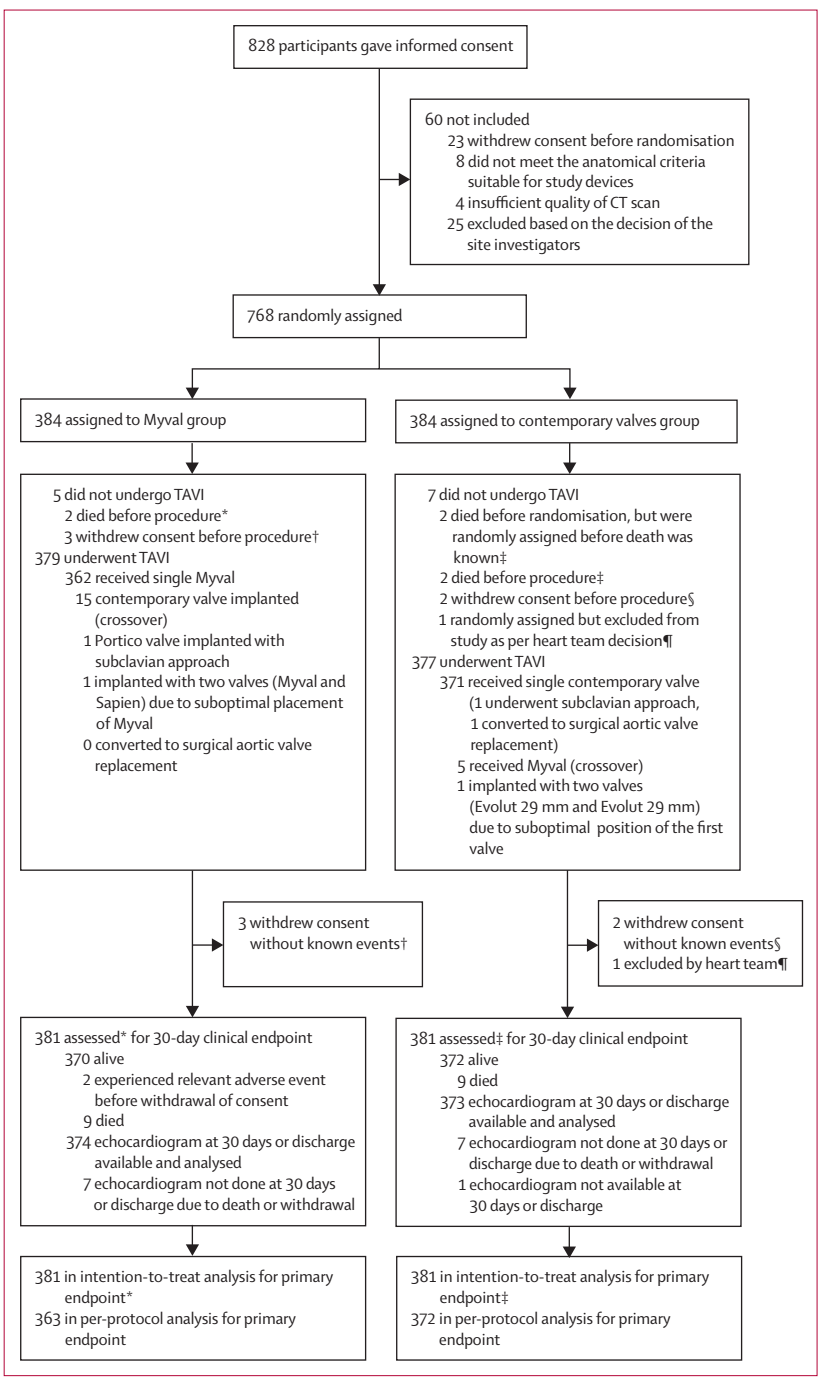


Figure 1: **Trial profile**
TAVI=transcatheter aortic valve implantation. *Two participants died before the procedure and were included in the population for endpoint analysis. †Three participants who withdrew consent before the procedure without any known events at that point in time were excluded from the endpoint analysis. ‡Two participants who gave informed consent were randomly assigned but the investigators were unaware that they had died in the meantime, and they are included in the population for endpoint analysis. Two participants died before the procedure and were included in the population for endpoint analysis. §Two participants who withdrew consent before the procedure without any known events at that point in time were excluded from the endpoint analysis. ¶One participant was excluded after randomisation by the investigator due to rapid progression of Alzheimer's disease in the participant.

and 6-min walk test, and quality of life was assessed by a 12-item Short Form Survey. The present report includes the echocardiographic assessment of haemodynamic parameters at baseline, predischarge, and 30 days after the index procedure. For assessment of the prosthetic valve regurgitation, echocardiography was considered as the primary modality. Details regarding postdilatation, including the indication, were collected as part of the post-hoc analysis.

A masked, independent clinical events committee adjudicated the primary endpoint and composite secondary endpoints related to technical and device success according to VARC-3. Echocardiograms were analysed centrally by the CORRIB Core Laboratory (Galway, Ireland), and ECGs were analysed by the Cardiovascular European Research Centre (Paris, France). Imaging analysts were masked to allocated treatment group.

Statistical analysis

The sample size of 768 participants was calculated to show non-inferiority of the Myval THV group to the contemporary THV group in the intention-to-treat and per-protocol populations with a statistical power of 93% and a 5% level of significance, assuming an event rate of 26·10%[5,6,20,21,30–33] in both groups and a non-inferiority margin of 10·44%. Although a sample size of 692 participants (346 in each group) was adequate for the intention-to-treat population, an attrition rate of 10% was assumed against each group. Therefore, 384 participants were needed in each group, giving a sample size of 768. All randomly assigned participants were evaluated for safety and effectiveness of the assigned device. The primary endpoint analysis was conducted in the intention-to-treat population. The composite secondary endpoints were analysed in the intention-to-treat, per-protocol, and as-treated populations. For non-inferiority assessment of the primary endpoint, a one-sided 95% CI was calculated using the Farrington–Manning test. Analyses in the intention-to-treat population were considered primary and those in the per-protocol population were considered supportive. Clinical events were censored at 30 days after the procedure. Deaths occurring before the procedure, but after randomisation, were considered to have occurred on day 1.

The demographic and baseline characteristics were summarised using descriptive statistics. For continuous variables, summary statistics included mean (SD) and median (IQR), as appropriate. One-way ANOVA or the Kruskal–Wallis test was used to compare continuous variables. Pearson's χ^2 test or Fisher's exact test was used to compare categorical variables as appropriate; categorical variables are presented as frequency and percentage. The mean difference and risk ratio of the two groups are presented with 95% CIs. ANOVA was used to compare the three types of devices, whereas

post-hoc analyses using Bonferroni correction were performed for pairwise comparison. ANCOVA was used with adjustment for baseline EOA. An independent statistician, who was masked to treatment allocation until unmasking, performed the data analysis. Statistical analysis was performed using R (version 4.3.3). A p value of less than 0·05 was considered significant.

An independent data and safety monitoring committee oversaw the safety of all participants.

Role of the funding source

The funder provided all financial support to conduct this trial and designed the trial protocol in consultation with the steering committee, but had no role in data analysis, data interpretation, or writing of the report.

Results

Between Jan 6, 2021, and Dec 5, 2023, 768 participants with severe symptomatic native aortic stenosis were recruited, with 384 participants randomly assigned to the Myval THV group and 384 to the contemporary THV group (figure 1). There were 768 participants in the intention-to-treat population and 735 in the per-protocol population. One participant was randomly assigned to receive Myval THV but was implanted with an XL size of Myval (30·5 mm) at the investigator's discretion (appendix 2 pp 9–10). This procedure constitutes a protocol violation, but the participant was kept in the intention-to-treat population. For the primary endpoints at 30 days, no imputation technique was used for missing outcomes. Only six (<1%) participants had missing 30-day outcomes from the study: five due to withdrawals by participants and one due to withdrawal decided by the operator because of a fast mental deterioration of the participant.

117 (15%) enrolled participants were younger than 75 years of age and 651 (85%) were aged 75 years or older. The mean age was 80·0 years (SD 5·7) in the Myval THV group and 80·4 years (5·4) in the contemporary THV group. The median STS score was 2·6% (IQR 1·7–4·0) in the Myval THV group and 2·6% (1·7–4·0) in the contemporary THV group. Both groups were similar in terms of baseline demographic characteristics and medical history (table 1).

In the Myval group, 379 (99%) participants underwent TAVI and 362 (94%) received a single Myval THV. One participant received a Myval THV and a comparator. 15 (4%) participants received the comparator instead of Myval, and one participant received a valve that was not included in the trial (Portico). In the contemporary group, 377 (98%) participants underwent TAVI and 371 (97%) received a single contemporary valve. Five (1%) participants in the contemporary group received Myval. One participant received two Evolut (figure 1). An inventory of the valves implanted in each group, including crossover to the comparator or other valve, shows that

	Myval THV series (n=384)	Contemporary THV series (n=384)
Age, years	80·0 (5·7)	80·4 (5·4)
Sex		
Female	193 (50%)	176 (46%)
Male	191 (50%)	208 (54%)
BMI, kg/m²	28·2 (4·9)	28·0 (4·9)
Symptoms		
Syncope not related to atrioventricular block	24 (6%)	26 (7%)
Hospitalisation for symptoms of aortic stenosis in the last 6 months	54 (14%)	69 (18%)
New York Heart Association classification III or IV	206/383 (54%)	196/383 (51%)
Canadian Cardiovascular Society grade III or IV	30/378 (8%)	26/371 (7%)
STS Predicted Risk of Mortality, %	2·6% (1·7–4·0)	2·6% (1·7–4·0)
Low score (<4)	290 (76%)	289 (75%)
Intermediate score (4–8)	78 (20%)	78 (20%)
High score (>8)	16 (4%)	17 (4%)
Medical conditions and medical history		
Diabetes	111 (29%)	114 (30%)
Hypercholesterolaemia	42 (11%)	36 (9%)
Hypertension	256 (67%)	254 (66%)
Current smoker	104 (27%)	86 (22%)
Alcohol consumption	89 (23%)	78 (20%)
Coronary artery disease	55 (14%)	58 (15%)
Chronic obstructive pulmonary disease	42 (11%)	40 (10%)
Estimated glomerular filtration rate <60 mL/min	171/362 (47%)	176/360 (49%)
Estimated glomerular filtration rate <30 mL/min	53/362 (15%)	54/360 (15%)
History of atrial fibrillation or flutter	94 (24%)	99 (26%)
Permanent pacemaker	11 (3%)	18 (5%)
Previous myocardial infarction	26 (7%)	23 (6%)
Previous PCI	30 (8%)	25 (7%)
Previous CABG	13 (3%)	21 (5%)
Previous aortic valvuloplasty	4 (1%)	0
Previous stroke	13 (3%)	8 (2%)
Cerebrovascular accident or transient ischaemic attack in last 6 months	5 (1%)	1 (<1%)
Echocardiography findings		
Aortic valve mean gradient, mm Hg	39·9 (14·0) (n=368)	38·7 (13·6) (n=368)
Aortic valve peak gradient, mm Hg	65·2 (21·6) (n=368)	63·0 (20·6) (n=368)
Aortic valve peak velocity, m/s	4·0 (0·7) (n=368)	3·9 (0·6) (n=368)
Aortic valve area, cm²	0·7 (0·2) (n=364)	0·7 (0·2) (n=360)
Left ventricular ejection fraction, %	58·3% (10·7) (n=349)	57·3% (10·0) (n=355)
Systolic pulmonary artery pressure, mm Hg	34·0 (15·0) (n=289)	33·9 (12·7) (n=296)
Moderate or higher aortic regurgitation	42/371 (11%)	47/374 (13%)
Moderate or higher mitral regurgitation	47/370 (13%)	51/375 (14%)
Moderate or higher tricuspid regurgitation	37/370 (10%)	44/375 (12%)

(Table 1 continues on next page)

	Myval THV series (n=384)	Contemporary THV series (n=384)
(Continued from previous page)		
CT findings		
Aortic annulus perimeter, mm	77·8 (6·7)	77·9 (6·5)
Aortic annulus area, mm²	470·5 (80·0)	471·4 (78·4)
Small annulus (aortic annulus area ≤430 mm²)	125 (33%)	120 (31%)
Total calcium aortic valve, mm³	845·1 (540·8–1310·0)	835·0 (476·0–1295·9)
Bicuspid valve	23 (6%)	29 (8%)
Type 0	5/23 (22%)	0
Type 1a	13/23 (57%)	24/29 (83%)
Type 1b	5/23 (22%)	5/29 (17%)

Data are mean (SD), n (%), n/N (%), or median (IQR). THV=transcatheter heart valve. STS=Society of Thoracic Surgeons. PCI=percutaneous coronary intervention. CABG=coronary artery bypass graft.

Table 1: **Baseline demographics, medical history, and preprocedural imaging**

See Online for appendix 1
See Online for appendix 2

48% of Myval THV implants were intermediate-sized devices (appendix 2 pp 9–10).

Postdilatation was performed significantly less frequently with the Myval THV than with the comparators (10% *vs* 21%, p<0·0001; table 2, appendix 2 p 11). Quantitative videodensitometry using the final aortogram showed significantly less regurgitation (median 3·0 [IQR 1·0–7·0] *vs* 4·0 [1·0–9·0], p=0·049), and a significantly lower rate of regurgitation fraction above 17% (2·0% *vs* 6·0%, p= 0·025) in the Myval compared with the contemporary THV group.

At 30 days, the primary composite endpoint occurred in 94 (25%) of 381 participants in the Myval group and 103 (27%) of 381 in the contemporary THV group (risk difference –2·3%). The upper limit of the 95% CI for the difference in the primary endpoint was 3·8% and, given the predefined non-inferiority margin was 10·44%, the Myval THV achieved non-inferiority compared with the contemporary THV with statistical significance ($p_{non\text{-}inferiority}$<0·0001; figure 2A). Secondary analyses of the components of the primary endpoint showed no significant between-group differences (figure 2B, table 3). In the intention-to-treat population there were no significant between-group differences in secondary endpoints of rates of technical success, 30-day device success, and early safety (table 4, appendix 2 pp 12–13).

Preprocedural aortic annulus areas derived from MSCT, and EOAs on follow-up echocardiography (at 30 days or discharge) were compared according to device type and size (appendix 2 pp 14–16). The intermediate sizes were only available for Myval and not for the contemporary devices. EOA at follow-up was statistically significantly better for the 23 mm, 26 mm, and 29 mm Myval compared with Sapien, and there were no notable differences in the preprocedural annular area. BEV and SEV sizing recommendations vary between manufacturers and, thus, preprocedural annular area and EOA at follow-up for the same valve diameters might not always be comparable. Overall, the significantly larger EOA after TAVI with Myval can be attributed to the availability of the intermediate sizes (figure 3, appendix 2 pp 14–16), which were implanted in 177 (48%) of 368 participants in the participants treated with Myval. At 30-day follow-up, 19 (5%) participants in the Myval group and 20 (5%) in the contemporary group did not have echocardiograms available at 30 days and their predischarge echo served as a substitute. Except for participants who died or withdrew consent, echocardiograms either at 30 days or at predischarge were available and analysable in all participants but one, who did not undergo any echocardiography after TAVI (figure 1).

No significant between-group differences were seen for changes in the duration of the PR and QRS intervals between preassessment, predischarge post implant, and 30-day follow-up (appendix 2 p 17). The incidences of new PPI (Myval THV 57 [15%] of 381 *vs* contemporary THV 65 [17%] of 381, p=0·49; figure 2B, table 3) and left bundle branch block (Myval THV 39 [12%] of 339 *vs* contemporary THV 40 [12%] of 331, p=0·91; appendix 2 p 17) at 30 days were also similar.

In both groups there were significant functional improvements from baseline to 30-day follow-up as assessed by New York Heart Association functional assessment (appendix 2 p 18) and the 6-min walk test (appendix 2 p 19). Quality of life, as assessed by the 12-item Short Form Survey, also significantly improved in both groups from baseline to 30-day follow-up, with improvements seen in both physical and mental health components (appendix 2 p 20).

Discussion

The principal findings at 30-day follow-up of the randomised LANDMARK trial among individuals with symptomatic native severe aortic stenosis treated with TAVI are: that the Myval THV was proven to be non-inferior to two contemporary THVs for the primary composite endpoint of safety and effectiveness at 30 days (25% *vs* 27%); that when itemised, there were no significant between-group differences in any components of the primary composite endpoint; that in both groups there were similar significant improvements in haemodynamic parameters, mean pressure gradients, and effective orifice areas between baseline and 30 days (figure 3); and that in more than 48% of participants treated with Myval, the intermediate sizes were selected according to the quantitative measurement of the aortic annulus area provided by preprocedural MSCT.

Since 2020, there have been three head-to-head comparisons of contemporary TAVI devices in an effort to help interventionalists select the most appropriate device for their patients.[22–24] The SCOPE II trial compared two SEVs (ACURATE neo and CoreValve Evolut) and reported a higher incidence of all-cause death (3% *vs* 1%) and stroke (4% *vs* 3%) at 30 days with ACURATE neo.

	Myval THV series (n=384)	Contemporary THV series (n=384)	Mean difference or risk difference	p value
TAVI procedure not attempted	5 (1%)	7 (2%)	−0·5% (−2·5 to 1·5)	0·77
Randomly assigned after death*	0	2 (<1%)	−0·5% (−1·5 to 0·5)	0·50
Death after randomisation before procedure	2 (<1%)	2 (<1%)	0·0% (−1·0 to 1·0)	1·00
Consent withdrawn before procedure	3 (<1%)	2 (<1%)	0·3% (−1·1 to 1·7)	1·00
Patient withdrawn by heart team before procedure	0	1 (<1%)	−0·3% (−1·0 to 0·5)	1·00
TAVI procedure performed				
Transfemoral approach	378/379 (>99%)	376/377 (>99%)	0·0% (−0·7 to 0·7)	1·00
Percutaneous	337/379 (89%)	337/377 (89%)	−0·5% (−5·2 to 4·2)	0·93
Surgical cutdown	41 (11%)	39 (10%)	0·5% (−4·2 to 5·1)	0·93
Trans-subclavian approach	1 (<1%)	1 (<1%)	0% (−0·5 to 1·0)	1·00
Percutaneous	0	0	..	..
Surgical cutdown	1 (<1%)	1 (<1%)	0% (−0·7 to 0·7)	1·00
Mean procedural time, min	77·0 (40·3)	77·6 (40·3)	−0·6 (−6·4 to −5·2)	0·75
Median procedural time, min	70·0 (49·0–96·5)	70·0 (50·0–95·0)	..	..
Mean total contrast volume, mL	143·6 (68·5)	149·8 (72·5)	−6·2 (−16·3 to −3·9)	0·29
Median total contrast volume, mL	140·0 (100·0–180·0)	135·0 (100·0–200·0)	..	..
General anaesthesia	73 (19%)	74 (20%)	−0·3% (−6·3 to 5·5)	0·97
Conscious sedation	306 (81%)	303 (80%)	0·3% (−5·5 to 6·3)	0·97
Cerebral protection device	48 (13%)	33 (9%)	3·9% (−0·8 to 8·6)	0·10
Predilatation	164 (43%)	144 (38%)	5·1% (−2·2 to 12·3)	0·18
TAVI device implanted	379	377	..	..
Median RF post implantation, before postdilatation, %	12·0% (6·0–18·5), n=23	12·0% (6·0–18·0), n=35	1·5% (0·7 to 3·6)	0·78
RF>17% post implantation, before postdilatation	6/23 (26%)	11/35 (31%)	−5·3% (−32·6 to 21·9)	0·88
Postdilatation	38 (10%)	80 (21%)	−11·2% (−16·6 to −5·8)	<0·0001
Median RF after postdilatation, %	2·0% (1·0–8·0), n=33	5·0% (1·0–9·0), n=64	0·8% (0·1 to 6·7)	0·19
RF>17% after postdilatation	0/33	5/64 (8%)	..	0·12
Median RF in final aortography, %	3·0% (1·0–7·0), n=295	4·0% (1·0–9·0), n=301	0·8% (0·1 to 5·5)	0·049
RF>17% in final aortography	6/295 (2%)	18/301 (6%)	−4·0% (−7·4 to −0·5)	0·025
Use of closure device	344 (91%)	340 (90%)	0·5% (−3·9 to 5·0)	0·89
Median length of index hospitalisation, days	4·0 (3·0–6·0)	4·0 (2·0–6·0)	0·7% (0·1 to 1·3)	0·014

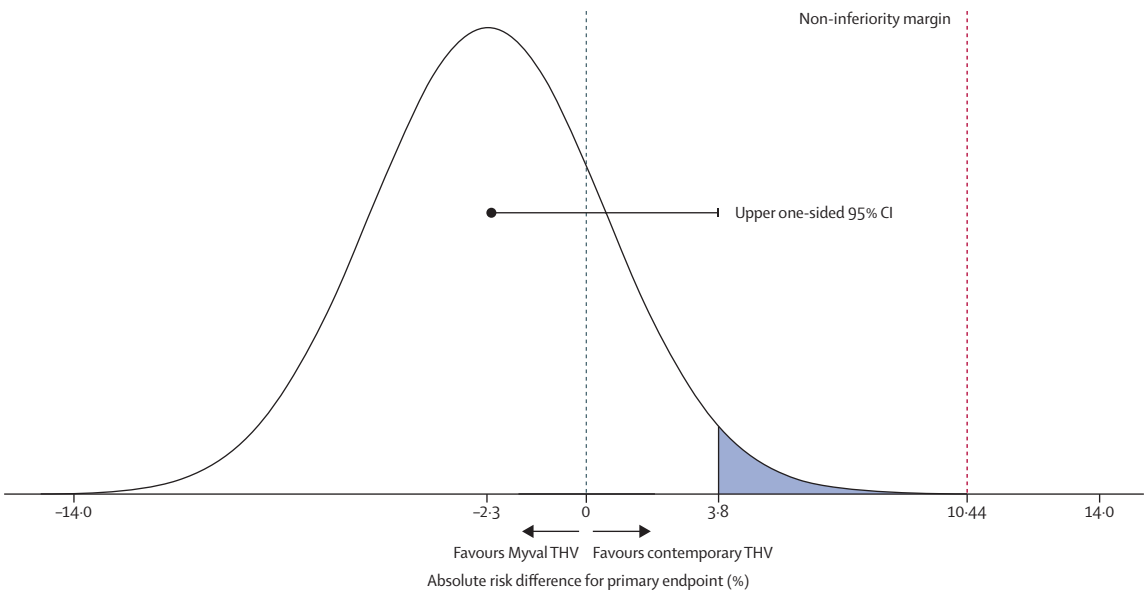
Data are n, n (%), n/N (%), mean difference or risk difference % (95% CI), mean (SD), median (IQR), or p value. THV=transcatheter heart valve. TAVI=transcatheter aortic valve implantation. RF=aortic regurgitation fraction on videodensitometric analysis of aortography. *Site was unaware of the deaths and randomly assigned these patients.

Table 2: **Procedural details**

Similarly to the results of the SCOPE I trial, ACURATE neo failed to meet the prespecified criteria for non-inferiority.[21,22] In the PORTICO IDE trial, the Portico valve (SEV, intra-annular) was compared with commercially available intra-annular BEVs (Edwards-Sapien, Sapien XT, or Sapien 3) or supra-annular SEVs (CoreValve, Evolut-R, or Evolut-PRO). At 30 days, the primary safety endpoint occurred more frequently in the Portico valve group compared with the commercial THV group (52 [14%] *vs* 35 [10%]; absolute difference 4·2 [95% CI −0·4 to 8·8, upper confidence bound 8·1%]; $p_{non\text{-}inferiority}$=0·034, $p_{superiority}$=0·071), as did new PPIs (27% *vs* 12%).[24] SOLVE-TAVI randomly assigned 447 participants with severe aortic stenosis undergoing TAVI to the Evolut R SEV or Sapien 3 BEV. At 30 days, both valves met the criteria of equivalence, with rates of meeting the composite primary efficacy endpoint of 28% for the SEV and 26% for the BEV. All-cause mortality (3% *vs* 2%), PPI (23% *vs* 19%), and paravalvular leaks (3% *vs* 2%) were numerically higher in the SEV group, whereas stroke (5% *vs* <1%) was numerically higher in the BEV group.[23]

The average STS in previous trials randomising TAVI devices was higher than 4% (with the exception of the SCOPE I trial), compared with the median in the current LANDMARK trial of 2·6%.[9,20–24] This difference reflects broadening of the indications for TAVI, albeit including individuals with greater anatomical complexities; the frequency of a bicuspid aortic valve was 7% and that of a small annulus (≤430 mm²) was 32% in the current cohort.

The primary composite endpoint at 30 days was adjudicated according to the new VARC-3 consensus,



B

	Events, n/N (%)		Risk difference, % (95% CI)	p value
	Myval THV series (n=384)	Contemporary valves (n=384)		
Primary analysis				
Primary endpoint (non-inferiority analysis)	94/381 (25%)	103/381 (27%)	–2·3% (NA to 3·8)	<0·0001
Secondary analysis				
Primary endpoint (superiority analysis)	94/381 (25%)	103/381 (27%)	–2·3% (–8·8 to 4·1)	0·51
Individual components of primary endpoint				
All-cause mortality	9/381 (2%)	9/381 (2%)	0% (–2·2 to 2·2)	1·00
All stroke	12/381 (3%)	12/381 (3%)	0% (–2·5 to 2·5)	1·00
Bleeding (type 3 and type 4)	11/381 (3%)	5/381 (1%)	1·6% (–0·7 to 3·9)	0·21
Acute kidney injury (stage 2, stage 3, and stage 4)	6/381 (2%)	3/381 (1%)	0·8% (–1·0 to 2·6)	0·50
Moderate or severe prosthetic valve regurgitation	13/381 (3%)	17/381 (4%)	–1·1% (–4·1 to 2·0)	0·58
Conduction system disturbances resulting in a new permanent pacemaker	57/381 (15%)	65/381 (17%)	–2·1% (–7·6 to 3·4)	0·49
Major vascular complications	6/381 (2%)	9/381 (2%)	–0·8% (–3·0 to 1·4)	0·60

Figure 2: **Risk difference for the primary endpoint**
(A) Probability distribution of the risk difference for frequency of the primary endpoint between the Myval and contemporary groups. The graph shows non-inferiority of the Myval THV series as the upper limit of the 95% CI (3·8) is less than the boundary of the non-inferiority margin (10·44). (B) Primary and secondary analyses of the primary endpoint and its components in the intention-to-treat population. The red line indicates the non-inferiority margin (prespecified at 10·44%). All 95% CIs and p values are two-sided except those of the primary, non-inferiority analysis (one-sided). THV=transcatheter heart valve.

which consists of seven endpoints including the incidence of PPI. The overall composite event rate is therefore accrued compared with previous head-to-head randomised studies; however, reassuringly, rates of individual components were similar. For example, rates of PPI in the present study were 15% in the Myval group and 17% in the contemporary group, compared with 19% in the Sapien 3 group and 23% in the Evolut R group of the SOLVE-TAVI trial.[23] In the SCOPE I trial, the rate of PPI was 9% for Sapien; in the SCOPE 2 trial, it was 18% for Evolut; and in the PORTICO IDE trial, it was 12% in the contemporary group.[21,22,24] In the study by Santos-Martinez and colleagues,[34] the rates of PPI were 7% for Myval, 13% for Sapien 3, and 19% for Evolut, whereas the respective increases in QRS interval were 11·0 ms, 14·5 ms, and 21·3 ms. In the LANDMARK trial, the increases in QRS interval at discharge were 15·4 ms for the Myval group and 16·8 ms for the contemporary group.

	Myval THV series (n=384)	Contemporary THV series (n=384)	Risk difference*	p value*
Primary endpoint at 30 days (non-inferiority analysis)	94/381 (25%)	103/381 (27%)	−2·3% (NA to 3·8)†	<0·0001†
All-cause mortality	9/381 (2%)	9/381 (2%)	0% (−2·2 to 2·2)	1·00
All stroke	12/381 (3%)	12/381 (3%)	0% (−2·5 to 2·5)	1·00
Bleeding (types 3 and 4)	11/381 (3%)	5/381 (1%)	1·6% (−0·7 to 3·9)	0·21
Acute kidney injury (stages 2–4)	6/381 (2%)	3/381 (<1%)	0·8% (−1·0 to 2·6)	0·50
Moderate or severe prosthetic valve regurgitation	13/381 (3%)	17/381 (5%)	−1·1% (−4·1 to 2·0)	0·58
Conduction system disturbances resulting in a new PPI	57/381 (15%)	65/381 (17%)	−2·1% (−7·6 to 3·4)	0·49
Major vascular complications	6/381 (2%)	9/381 (2%)	−0·8% (−3·0 to 1·4)	0·60
Secondary endpoints at 30 days				
All-cause mortality	9 (2%)	9 (2%)	0% (−2·2 to 2·2)	1·00
Cardiovascular mortality	9 (2%)	9 (2%)	0% (−2·2 to 2·2)	1·00
Valve-related mortality	0	1 (<1%)	−0·3% (−1·0 to 0·5)	1·00
Non-cardiac death	0	0	..	..
Stroke	12 (3%)	12 (3%)	0% (−2·5 to 2·5)	1·00
Fatal stroke	3 (<1%)	1 (<1%)	0·5% (−0·8 to 1·8)	0·62
Disabling stroke	5 (1%)	4 (1%)	0·2% (−1·5 to 2·1)	1·00
Non-disabling stroke	4 (1%)	7 (2%)	−0·8 (−2·7 to 1·2)	0·54
Acute kidney injury (stages 2–4)	6 (2%)	3 (<1%)	0·8% (−1·0 to 2·6)	0·51
Acute kidney injury stage 4	1 (<1%)	1 (<1%)	0% (−0·7 to 0·7)	1·00
Acute kidney injury stage 3	3 (<1%)	1 (<1%)	0·5% (−0·8 to 1·8)	0·62
Acute kidney injury stage 2	2 (<1%)	1 (<1%)	0·2% (−0·9 to 1·4)	1·00
Prosthetic valve regurgitation (moderate and severe)	13 (3%)	17 (5%)	−1·1% (−4·1 to 2·0)	0·58
Prosthetic valve regurgitation (moderate)	12 (3%)	16 (4%)	−1·1% (−4·0 to 1·9)	0·56
Prosthetic valve regurgitation (severe)	1 (<1%)	1 (<1%)	0% (−0·7 to 0·7)	1·00
Conversion to open surgery: sternotomy or thoracotomy with cardiopulmonary bypass	0	1 (<1%)	−0·3% (−1·0 to 0·5)	1·00
Sternotomy or thoracotomy without cardiopulmonary bypass	2 (<1%)	0	0·5% (−0·5 to 1·5)	0·50
Implantation of multiple (>1) transcatheter valves during the index hospitalisation	2 (<1%)	1 (<1%)	0·2% (−0·9 to 1·4)	1·00
TAV in TAV deployment	1 (<1%)	1 (<1%)	0% (−0·7 to 0·7)	1·00
Valve malposition	2 (<1%)	3 (<1%)	−0·3% (−1·7 to 1·2)	1·00
Hospitalisation for valve-related symptoms or worsening congestive heart failure	12 (3%)	18 (5%)	−1·6% (−4·7 to 1·4)	0·34
Severe Patient Prosthesis Mismatch (as defined by VARC-3)‡	15/372 (4%)	14/367 (4%)	0·2% (−2·8 to 3·2)	1·00
Incidence of patients with mean gradient >20 mm Hg	3/355 (<1%)	4/352 (1%)	−0·3% (−2·0 to 1·5)	0·72

Data are n (%), n/N (%), risk difference % (95% CI), or p value. THV=transcatheter heart valve. PPI=permanent pacemaker implantation. TAV=transcatheter aortic valve implantation. VARC-3=third Valve Academic Research Consortium. iEOA=indexed effective orifice area. *All 95% CIs and p values are two-sided except those of the primary composite endpoint analysis for non-inferiority (one-sided). †The 95% CI and p value are for non-inferiority. ‡If BMI <30 kg/m², iEOA ≤0·65. If BMI ≥30 kg/m², iEOA ≤0·55.

Table 3: **Study endpoints**

The rates of all stroke were 3% in both the Myval and contemporary groups, and the rates of disabling stroke were both 1% (table 3). SOLVE-TAVI reported a <1% all-stroke rate for Evolut R and a 5% rate for Sapien 3.[23] In the SCOPE 2 trial, the rate of all stroke was 4% in the CoreValve Evolut group, which included 3% of disabling stroke.[22] The rate of all stroke was 3% and that of disabling stroke was 1% in the commercially available group of the PORTICO IDE trial.[24]

The novel and salient feature of the Myval THV is its wide range of nominal device sizes, with 1·5 mm increments in diameter providing intermediate nominal valve diameters (21·5 mm, 24·5 mm, 27·5 mm, and extra-large 30·5 mm and 32 mm) in addition to conventional sizes (appendix 2 pp 9–10, 14–15). The intermediate sizes rely on manufacturing the cobalt alloy tube in the different specific diameters before laser cutting the valve frame.

Diversifying device sizes enables a more precise and appropriate match between the device and the individual's aortic annular dimensions. Mismatch between the two could result in oversizing the device with respect to the individual's anatomy, resulting in aortic annular rupture or conduction disturbances, whereas undersizing could lead to paravalvular leaks, malapposition, or even valve migration. Severe patient–prosthesis mismatch was 4% in both the Myval and contemporary groups (table 3).

	Myval THV series (n=379)	Contemporary THV series (n=377)	Risk difference	p value
Technical success (VARC-3) at exit from procedure room	365 (96%)	365 (97%)	−0·5% (−3·4 to 2·4)	0·85
Mortality	0	0	..	..
Unsuccessful access, unsuccessful delivery of the device, or unsuccessful retrieval of the delivery system	6 (2%)	2 (<1%)	1·1% (−0·7 to 2·8)	0·29
Malpositioning of valve	2 (<1%)	3 (<1%)	−0·3% (−1·7 to 1·2)	0·69
Multiple valve implantation	2 (<1%)	1 (<1%)	0·2% (−0·9 to 1·4)	1·00
Surgery or intervention related to the device or to a major vascular, access-related, or cardiac structural complication	10 (3%)	9 (2%)	0·2% (−2·3 to 2·7)	1·00
Sternotomy or thoracotomy with cardiopulmonary bypass (conversion to surgical valve)	0	1 (<1%)	−0·3% (−1·1 to 0·5)	0·50
Sternotomy or thoracotomy without cardiopulmonary bypass	2 (<1%)	0	0·5% (−0·5 to 1·5)	0·50
Vascular surgery related to the device	1 (<1%)	0	0·3% (−0·5 to 1·0)	1·00
Intervention related to the device	3 (<1%)	1 (<1%)	0·5% (−0·8 to 1·8)	0·62
Vascular surgery related to major vascular or access-related complications	2 (<1%)	3 (<1%)	−0·3% (−1·7 to 1·2)	0·69
Percutaneous intervention related to major vascular or access-related complications	1 (<1%)	4 (1%)	−0·8% (−2·2 to 0·6)	0·22
Percutaneous intervention related to cardiac structural complication	1 (<1%)	0	0·3% (−0·5 to 1·0)	1·00
Coronary artery obstruction requiring intervention	1 (<1%)	0	0·3% (−0·5 to 1·0)	1·00
Device success at 30 days	345 (91%)	338 (90%)	1·3% (−3·1 to 5·9)	0·61
Technical success (at the time of exit from the procedure room)	365 (96%)	365 (97%)	−0·5% (−3·4 to 2·4)	0·85
Mortality (after exit from the procedure room and before 30 days)*	7 (2%)	5 (1%)	0·5% (−1·5 to 2·6)	0·78
Surgery or intervention related to the device or to a major vascular, access-related, or cardiac structural complication (after exit from the procedure room and before 30 days)	2 (<1%)	0	0·5% (−0·5 to 1·5)	0·50
Vascular surgery related to major vascular or access-related complications	1 (<1%)	0	0·3% (−0·5 to 1·0)	1·00
Percutaneous intervention related to major vascular or access-related complications	1 (<1%)	0	0·3% (−0·5 to 1·0)	1·00
Intended performance of the valve	18 (5%)	24 (6%)	−1·7% (−5·2 to 1·9)	0·42
Early safety at 30 days	286 (76%)	278 (74%)	1·8% (−4·8 to 8·2)	0·65

Data are n (%), risk difference % (95% CI), or p value. VARC-3=third Valve Academic Research Consortium. THV=transcatheter heart valve. TAVI=transcatheter aortic valve implantation. *All-cause mortality in those patients who underwent TAVI. Two participants in the Myval group died before the TAVI procedure; two participants in the contemporary group died before randomisation but were randomly assigned, the investigators being unaware of their deaths at the time of randomisation; and two participants died before the TAVI procedure.

Table 4: **Technical success at exit from procedure room and device success and early safety at 30 days (VARC-3)**

The rates of predilatation in the present study were 43% in the Myval group and 38% in the contemporary group. In the PORTICO IDE trial, the rate of predilatation was 55% for the contemporary group.[24] The rate of predilatation for Sapien 3 was 23% in the SCOPE I trial, and for Evolut it was 41% in the SCOPE II trial.[21,22]

The relatively greater need for postdilatation in the contemporary THV group (80 [21%] participants) is striking when compared with Myval (38 [10%] participants, p<0·0001), especially as the Myval group also had a significantly lower incidence of unfavourable final regurgitation fraction (2% *vs* 6%, p=0·025). However, this difference in postdilatation was essentially driven by the Evolut SEV group, which had rates of postdilatation of 33%, compared with the similar rates of 10% for the Myval and Sapien BEV groups (χ^2 test p<0·0001). It is widely acknowledged that postdilatation following unsatisfactory valve implantation is not always a favourable intervention and could lead to detrimental outcomes.[35–38]

Hypoattenuated leaflet thickening was diagnosed in three participants on confirmatory postprocedural MSCT images obtained as per operator's discretion. For two of these participants, MSCT was performed due to the elevated transvalvular pressure gradient observed on echocardiogram. These two participants received oral anticoagulation therapy. For the other one participant, MSCT was performed to evaluate aortic dissection suspected on aortography. On day 15, the participant underwent surgical replacement of ascending aorta and THV, during which thrombi on the THV were confirmed. The THV was explanted and replaced with a 21 mm Magna surgical valve.

Beyond the clinical endpoints, extensive additional data have also been collected for each device, such as aortic annulus areas as measured on MSCT pre-implant, and EOA from echocardiograms post implant (appendix 2 pp 14–15). The long-term clinical implications of these observations, if any, are currently undetermined, but they will be investigated in the upcoming 10 years of

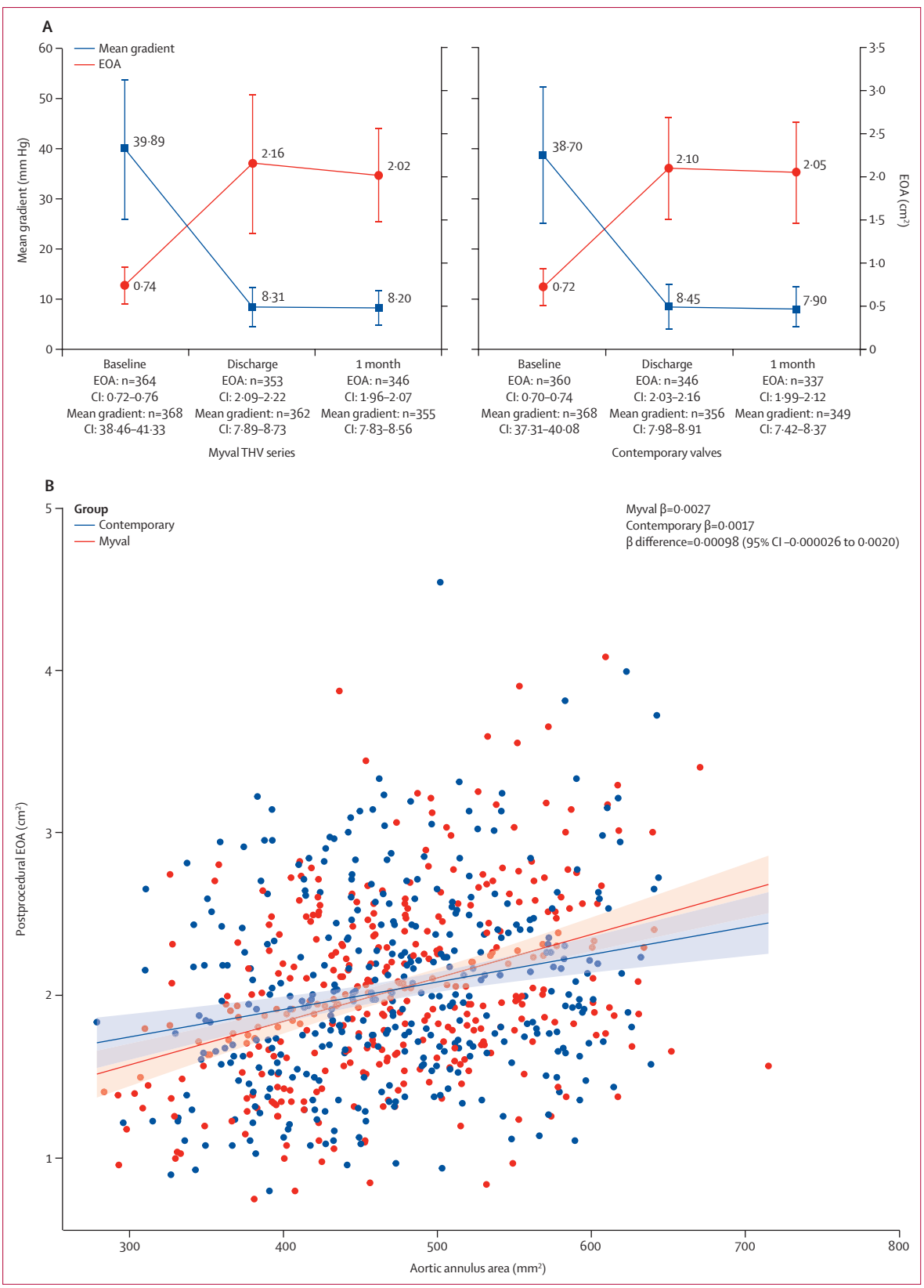


Figure 3: **Haemodynamic performance of Myval and contemporary THV series**
(A) EOA and mean pressure gradient at preprocedure, at discharge, and at 30-day follow-up. (B) Comparison of preprocedural aortic annulus area derived from multislice CT scan and the EOA on echocardiography at follow-up (30 days or discharge). β=coefficient of the slope of the linear regression. EOA=effective orifice area. THV=transcatheter heart valve.

planned and monitored follow-up. Different iterations of THVs were used in both groups and it might be argued that the benefit of the most recent generation of devices was not represented sufficiently in the trial. However, continuous evolution of technology is unavoidable and inherent to progress, and reflects recent and current daily clinical practice.

Although investigators were familiar with the implantation procedures of the two contemporary comparator valves, as Myval was a novel technology they were required to learn the techniques for implantation with its associated learning curve. Although the technical and device success rates were similar between the two groups, in the Myval group, annular rupture was observed in one participant and cardiac tamponade in two.

An imbalance exists in crossovers between the two groups (15 in Myval *vs* five in contemporary). For nine participants (eight Myval and one contemporary), crossover occurred for logistical reasons due to the allocated valve not being available in the required size. Nevertheless, the per-protocol analysis theoretically abrogates this imbalance. Due to the absence of masking including the masking of the participant, measurement bias was inherent to the intention-to-treat estimates. However, the clinical event committee was masked to the treatment group in order to minimise potential bias. We also acknowledge selection bias in the per-protocol estimates, confounding bias in the as-treated estimates, and selection bias due to missing outcome data. However, the frequency of missing outcomes was as low as 0·8%.

The indication of new PPI was left to the investigator's discretion, resulting in various rates of PPI among sites; however, this site heterogeneity did not affect the comparison between the two groups. In some centres, the indication for a PPI was decided after electrophysiological studies, whereas in others, the indication was based on ECG telemonitoring in the coronary care unit, with the decision made by the supervising cardiologist.

During the early phase of enrolment, recruitment was slow, primarily due to restrictions arising from the COVID-19 pandemic; however, litigation in some European countries also restricted the availability of Myval. The global inclusion rate of participants in this trial is 768 (15%) of 5109, and it is therefore plausible that before informed consent some investigators preselected participants for the trial, which potentially undermines its generalisability. Unfortunately, this is inherent in the design of a trial comparing multiple THVs; however, once randomly assigned, baseline characteristics of participants and diseased aortic valve characteristics were similar in both groups.

Finally, in the current report, follow-up duration was limited to 30 days to assess early safety and effectiveness as recommended by the VARC-3 consensus, and this does not permit any inference regarding long-term prognosis. Clinical and echocardiographic assessment, as well as monitoring of the collected data, will continue for the next 10 years.

In conclusion, the LANDMARK trial establishes the non-inferiority of Myval compared with contemporary THV for early safety and effectiveness composite endpoints at 30 days for the treatment of individuals with symptomatic severe native aortic stenosis. Primary selection of participants by the heart team has expanded the indications of TAVI to those with complex anatomy, such as small annuli and bicuspid aortic valves, as frequently encountered in clinical practice. The availability of intermediate sizes enables precise and appropriate matching of devices to an individual's anatomy; we acknowledge that the added value of the Myval THV in the current TAVI landscape, with its broader range of intermediate nominal sizes, remains hypothesis generating, and the hypothetical benefit of this tailored approach still needs to be investigated in future trials with specific predefined endpoints or by superior clinical outcomes in the long term. Continued follow-up over the next decade will further investigate long-term prognosis and valve durability.

Contributors

AB, YO, OS, and PWS conceptualised and supervised the study and were responsible for project administration. AB, M-CM, YO, OS, and PWS designed the methodology, validated the data, performed formal analysis, and were responsible for resources. NvR, IJA-S, MH, MB, AIj, PL, DU, BM, RSH, VN, MPr, BJWMR, PLM, FF, MDSA, EvB, AL, AIe, MM, MW, KT, ET, FB, MV, MPa, OA, W-KK, JR, IK, and VP conducted the investigation. AB, HE, M-CM, YO, OS, and PWS curated the data. PWS was responsible for funding acquisition. AB, HE, AT, M-CM, YO, OS, and PWS were responsible for software. AB and PWS visualised the data. AB, AT, YO, OS, and PWS wrote the original draft. All authors reviewed and edited the manuscript. PWS and the trial statistician had full access to all study data and had final responsibility for the decision to submit for publication.

Declaration of interests

NvR reports grant funding and personal fees from Abbott; grants from Philips, Biotronik, and Medtronic; and personal fees from MicroPort, Bayer, and RainMed Medical outside the submitted work. IJA-S reports being a proctor for Medtronic, Boston Scientific, and Meril Life Sciences. AIj reports consulting fees from Meril Life Sciences, Medtronic, Angiocare, Keystone Heart, PulseCath, Salveo, Abbott, Philips, Fysicon, Cardiawave, Pi Medical, and Svelte. DU reports lecture and educational event fees from Medtronic; a lecture fee from and being a proctor for Meril Life Sciences; and being a member of the Medtronic EMEA surgical advisory board. RSH reports grant funding from Bayer; and speaker fees from Amgen, Novartis, and Abbott outside the submitted work. MW reports institutional research funding from Emboline, Meril Life Sciences, Edwards Lifesciences, and TransAortic Medical. MPa reports honoraria from Boston Scientific, Philips, and Abbott. OA reports research grants and lecture fees from and being a proctor for Abbott; and advisory board participation for and support for attending meetings from Meril Life Sciences. W-KK reports honoraria or consultancy fees from Abbott, Boston Scientific, Edwards Lifesciences, Meril Life Sciences, and Hi-D Imaging; and participation on an advisory board or data and safety monitoring board for Abbott and Boston Scientific. JR reports being a consultant with Medtronic and Qatma; being a proctor for Medtronic; and travel support for attending meetings from Meril Life Sciences, Edwards Lifesciences, Abbott, Medtronic, and Boston Scientific. AT reports a grant from the Fukuda Foundation for Medical Technology. SG reports honoraria or consultancy fees from

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Data sharing

The study protocol is available in appendix 1. Other data, including study participant data, the statistical analysis plan, and informed consent, can be shared upon request to the corresponding author after the final long-term follow-up is published.

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